



STATS 141XP: Cry Detection Classification

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1 Introduction & Abstract

Background & Problem Statement

Background

- Can human babies' cries give information about their underlying cause?
- Understanding the emotions of a baby—tiredness, hunger, discomfort, or pain is importance in **communication** and **intervention**, especially because babies are unable to convey their needs and emotions through spoken language

Aim

- By analysing data about the acoustic features of baby cries, we aim to understand whether or not we can extract information about the cause of a baby cry.
- We train and test different models (including Logistic, KNN, Random Forest, and SVM) on our data to check classification accuracy

Introduction to the Data

1. Cry Detection Classifications

Description: df [6 × 3]

	file_name <chr>	Label <chr>	average_cry_strength <dbl>
1	02c3b725-26e4-4a2c-9336-04ddc58836d9-1430726196216-1.7-m-04-hu	Junk	NaN
2	02ead89b-aa02-453e-8b83-6ebde9fe7551-1430233132879-1.7-m-26-hu	Junk	NaN
3	035c6b30-a145-42b9-8d0f-445cd9003d2c-1435948197257-1.7-m-04-hu	Cry	1.0000000
4	03abcb8f-400a-47d8-ad82-7e4586cc06be-1431864192133-1.7-f-48-hu	Cry	1.0000000
5	03ADDCFB-354E-416D-BF32-260CF47F7060-1433658024-1.1-f-04-ti	Cry	1.0000000
6	045C5483-69E1-4BEC-B1D8-9286D174B9B2-1430102996-1.0-m-04-hu	Cry	0.6986819

6 rows

- 458 observations
- Predicts whether cry is actually a cry vs. junk noise

2. Acoustic Features Data

Description: df [458 × 6,376]

name <chr>	frameTime <dbl>	audspec_lengthL1norm_sma_range <dbl>
03ADDCFB-354E-416D-BF32-260CF47F7060-1433658024-1.1-f-04-ti	0	1.92960100
5B416CE8-2591-4531-9ADC-86D08585D48B-1430144827-1.0-m-48-ti	0	0.90935120
06c4cfa2-7fa6-4fda-91a1-ea186a4acc64-1430029221058-1.7-f-26-ti	0	0.29884550
06c4cfa2-7fa6-4fda-91a1-ea186a4acc64-1430029237378-1.7-f-26-ti	0	0.34928290
06c4cfa2-7fa6-4fda-91a1-ea186a4acc64-1430029246453-1.7-f-26-ti	0	1.50209600
7A22229D-06C2-4AAA-9674-DE5DF1906B3A-1436891944-1.1-m-72-ti	0	1.52202400
7A22229D-06C2-4AAA-9674-DE5DF1906B3A-1436891957-1.1-m-72-ti	0	1.63655000
79FF400A-97E2-4701-987D-C7C850D5523C-1430089487-1.0-f-48-ti	0	1.77661500
1309B82C-F146-46F0-A723-45345AFA6EA8-1430059864-1.0-f-04-ti	0	0.89743970
1309B82C-F146-46F0-A723-45345AFA6EA8-1431172241-1.0-f-48-ti	0	2.20595600

1-10 of 458 rows | 1-3 of 6376 columns

Previous 1 2 3 4 5 6 ... 46 Next

- 458 observations, 6373 acoustic features
- Gives information about acoustic features of audios of baby cries

Abstract



In this project, our research is focused on conducting a thorough review of existing literature by meticulously reviewing scholarly articles to identify the advantageous and disadvantageous aspects of various models, while simultaneously constructing both reliable and intentionally flawed models using the Baby Cry dataset, in other words, being separately good statisticians and bad statisticians. Among the effective statistical models, we employed techniques such as Logistic Regression, KNN, Random Forest, and SVM. All the methods were conducted twice as both using all classes and only two classes due to the domination of one type of cry. Notably, the Random Forest model recorded the highest test accuracy, achieving 85.71%. Conversely, being bad statisticians, we used the unfiltered data to conduct the same four models as mentioned above again. When modelling parameters on entire data (including test set), the Random Forest Method got the surprisingly high accuracy (100%). The following slides will discuss the details of model algorithms and challenges we faced during our project.



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Literature Review

Literature Review

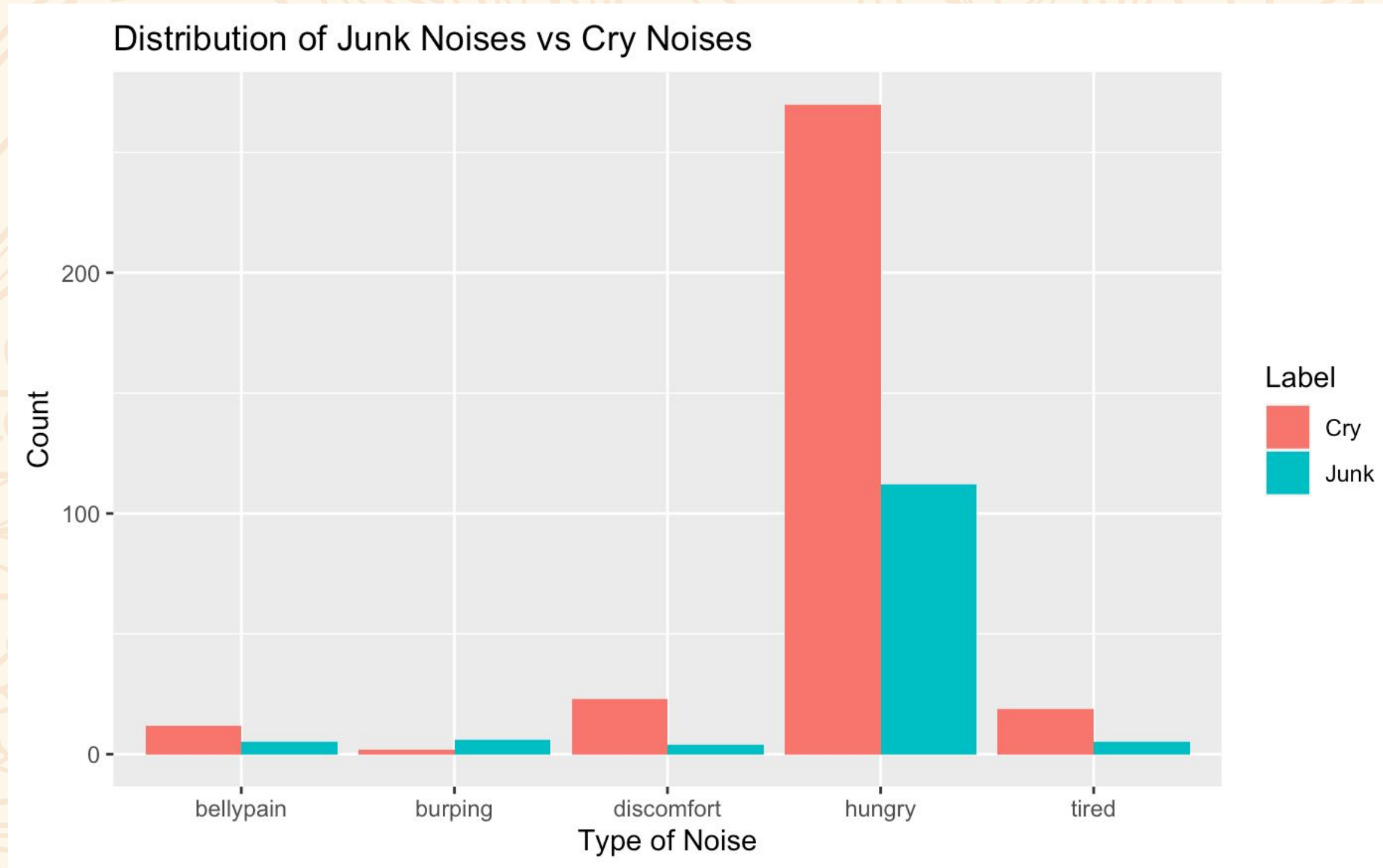
- Looked at a total of 7 research studies
- Majority use CNN models for classification
- Testing accuracy ranges from 81% - 100%%
- Very little acknowledgement of skewness of data:
- Some studies don't report cross-validation

study name and url	number of observations used	number of non-cries included in cry study	number of classes	which classes used	models used	training accuracy	testing accuracy	cross validation accuracy	other observations
Classifying Infant cry type (https://www.kaggle.com/code/kishoreharshankumar/classifying-infant-cry-type)				class_labels = ['belly pain','burping','discomfort','hungry','tired']	CNN	71.47%		72.53%	
Testing the Universal Baby Language Hypothesis - Automatic Infant Speech Recognition with CNNs (https://ieeexplore.ieee.org/document/8441412)	315 (250 small duration recordings for training, 65 for testing)	0	5	Neh" = hungry; "Eh" = need to burp; "Oah (Owh)" = tired (sleepy); "Eairh (Eargghh)" = stomach cramp (lower gas); "Heh" = physical discomfort at skin kevel (feeling hot or wet, for example)	CNN		89%		
train acc = 100% val acc ≈ 68% (Tensorflow) https://www.kaggle.com/code/philopateergeorgei/train-acc-100-val-acc-68-tensorflow			5	classes = ['belly_pain','burping','discomfort','hungry','tired']	CNN	67.39%	100%	67.39%	
Infant Weeping Calls Decoder using Statistical Feature Extraction and Gaussian Mixture Models (https://ieeexplore.ieee.org/abstract/document/8944527/references#references)	714		5	hunger, attention seek, lack of ease, stomach issue, unidentified	Gaussian Mixture Model		81.27%		
https://www.kaggle.com/code/menonbrur/cry-classification	Total: 457 Train set : 328 Validation set : 83 Test set : 46		5	CLASS_IDS = {"belly_pain": 0, "burping": 1, "discomfort": 2, "hungry": 3, "tired": 4}		84.75%	83%		
INFANT CRY ANALYSIS https://www.kaggle.com/code/jangambharadwaji/infant-cry-analysis			5	labels = {'belly_pain': 0, 'burping': 1, 'discomfort': 2, 'hungry': 3, 'tired': 4}	CNN				



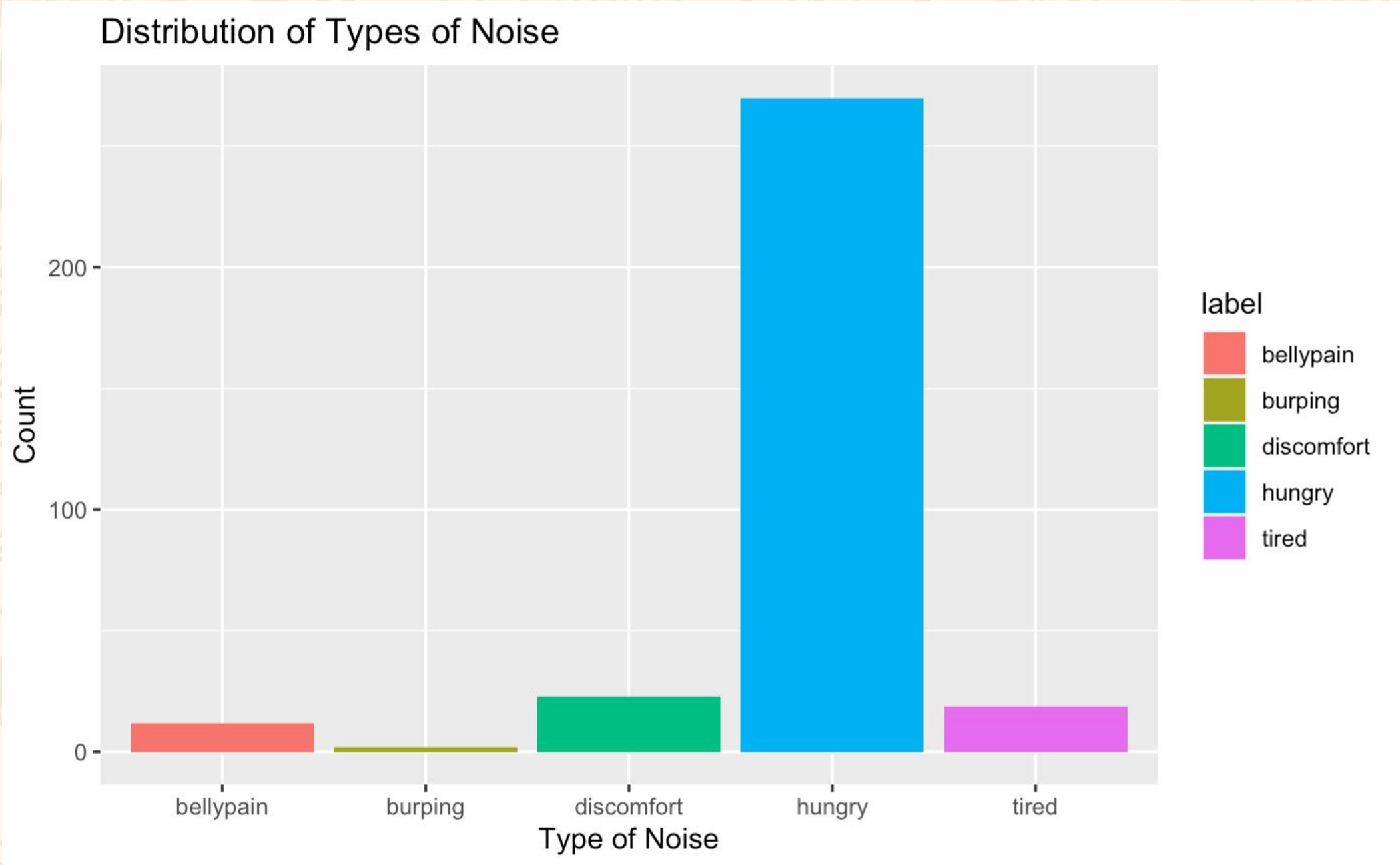
EDA & Methodology

Exploratory Data Analysis



- Outcome variable: prediction of cry noise with 5 classes
- Out of 458 observations:
 - 83% are classified as 'hungry' - 382 obs.
 - 326 classified as cry noises vs. 132 classified as junk noises

Exploratory Data Analysis - Filtering out Junk Noises



<u>Type of Cry</u>	<u>Count</u>
Belly	12
Discomfort	23
Hungry	270
Tired	19
Burping	2

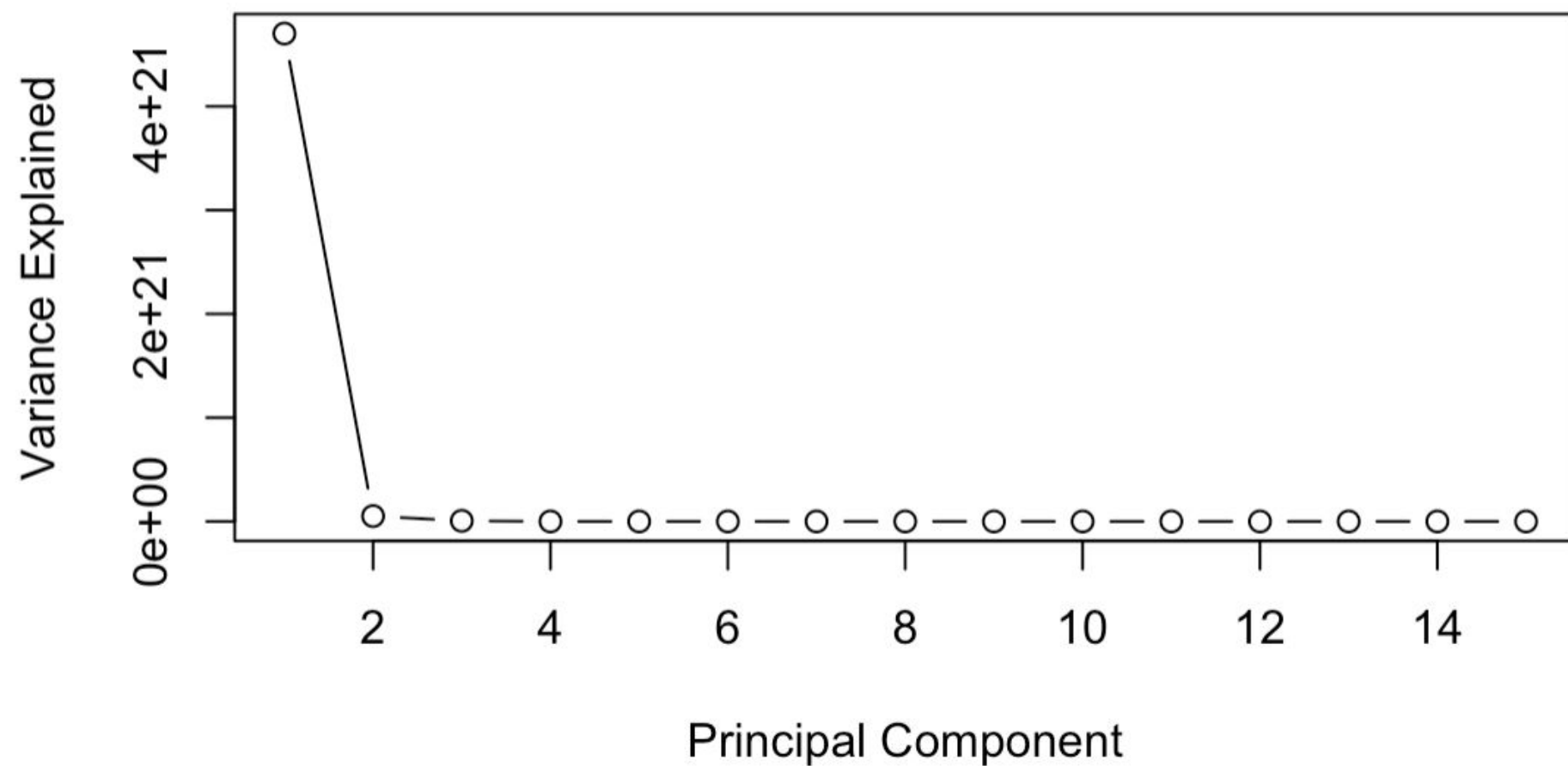
- 326 observations - majority are classified as hungry

Methodology

- Conduct PCA analysis on filtered data to find variables that contribute most to predicting the outcome variable
- Train and test 4 different models: Logistic Regression, K Nearest Neighbours, Random Forest, and Support Vector Machines (SVM)
- First train models to classify all 5 classes of types of cry & see prediction accuracy
- Next, since 'hungry' has the most number of observations, split data into 2 classes: 'hungry' & 'other', and train + test models again to check if prediction accuracy increases

Principal Component Analysis

Scree Plot



- 3 components that contributed the most to predicting the response variable



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Good Statistics: Models

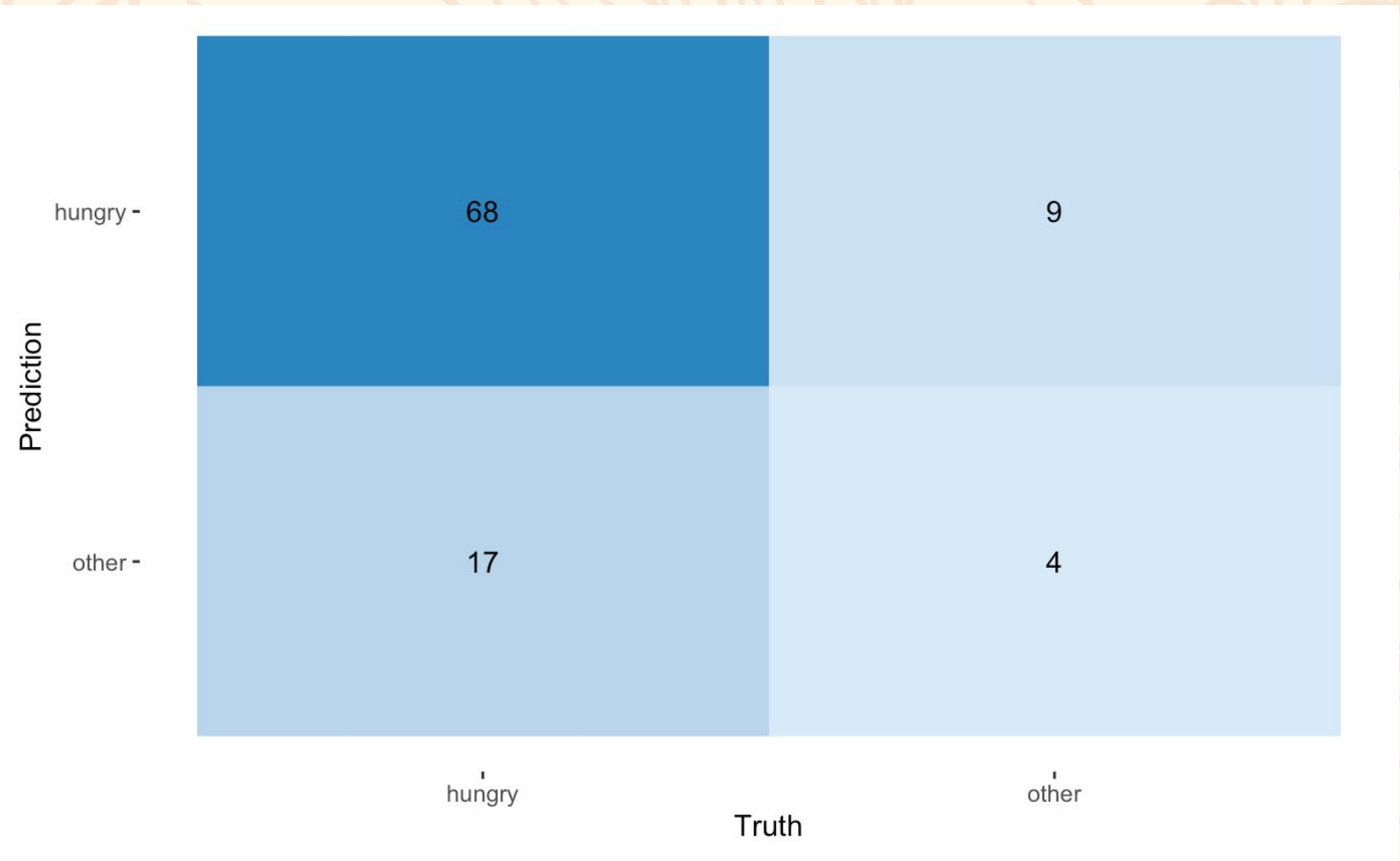
1. Logistic Regression - All Classes

Prediction					
	bellypain -	burping -	discomfort -	hungry -	tired -
	1	0	0	4	0
	1	0	0	6	1
	1	0	2	13	3
	0	0	4	27	1
Truth	0	0	0	31	0
	bellypain	burping	discomfort	hungry	tired

	Sensitivity	Specificity	Precision	Misclassification
Belly	20%	97.78%	33.33%	6.32%
Burping	0%	100%	NaN	8.42%
Discomfort	10.53%	94.74%	33.33%	22.11%
Hungry	84.38%	14.29%	33.33%	62.11%
Tired	0%	92.19%	0%	37.89%

Testing Accuracy: 31.58%
Kappa: -0.0042

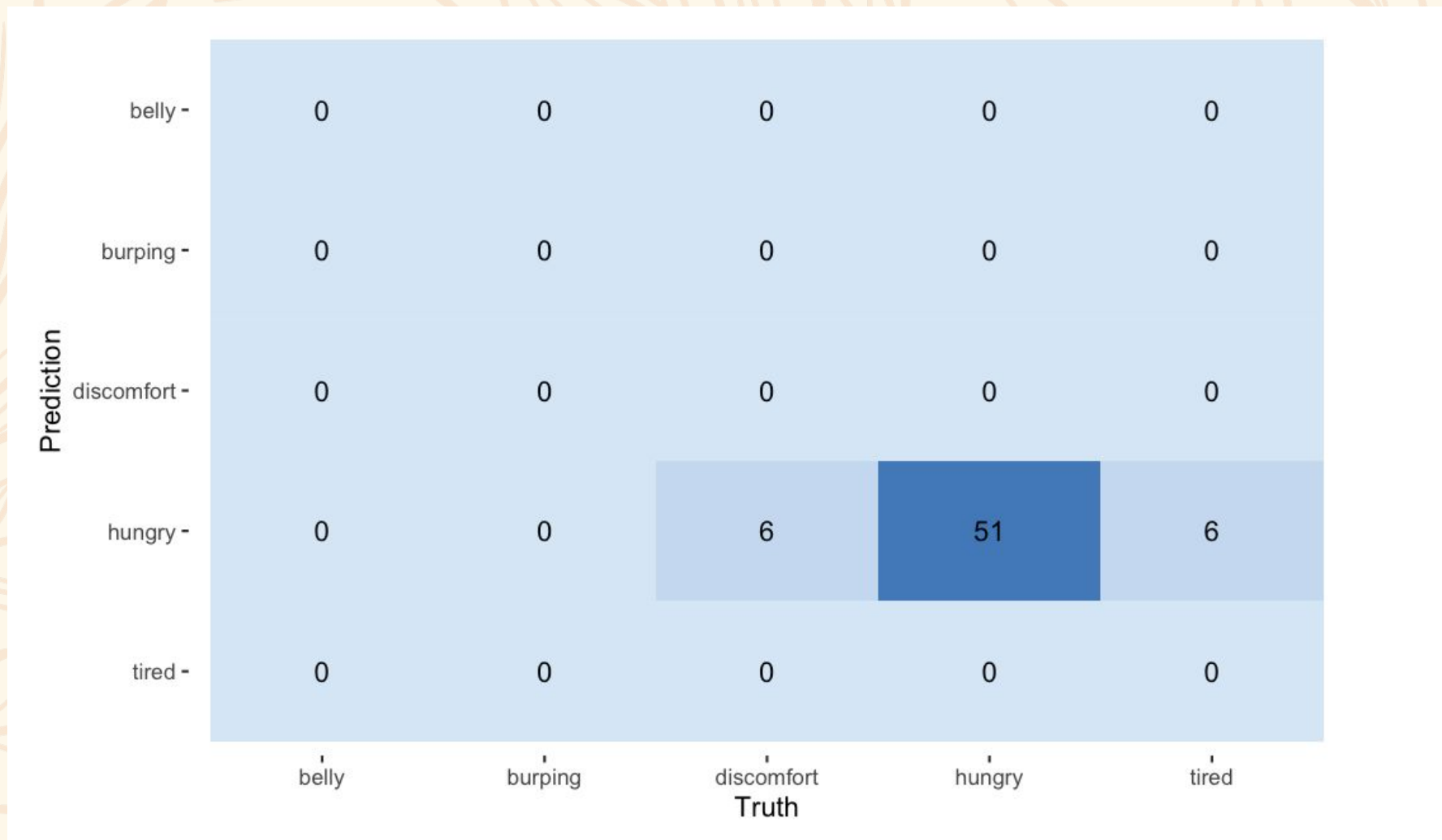
1. Logistic Regression - 2 Classes



Sensitivity	Specificity	Precision	Misclassification
88.31%	19.05%	80.00%	26.53%

Testing Accuracy: 73.47%
Kappa: 0.0854

2. K Nearest Neighbours (KNN) - All Classes



	Sensitivity	Specificity	Precision	Misclassification
Belly	NaN	100%	NaN	0%
Burping	NaN	100%	NaN	0%
Discomfort	0%	100%	NaN	9.52%
Hungry	100%	0%	80.95%	19.04
Tired	0%	100%	NaN	0.95%

Testing Accuracy: 80.95%
Kappa: 0

2. K Nearest Neighbours (KNN) - 2 Classes

Prediction	Truth	
	hungry	not hungry
hungry -	83	14
not hungry -	1	0

Sensitivity	Specificity	Precision	Misclassification
98.81	0%	85.57%	15.31%

Testing Accuracy: 84.69%
Kappa: -0.0194

3. Random Forest - All Classes

Prediction					
	bellypain -	burping -	discomfort -	hungry -	tired -
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	5	0	4	80	8
	Truth				
	bellypain	burping	discomfort	hungry	tired

	Sensitivity	Specificity	Precision	Misclassification
Belly	NaN	94.90%	0%	5.10%
Burping	NaN	100%	NaN	0%
Discomfort	NaN	95.92%	0%	4.08%
Hungry	82.47%	0%	98.77%	18.37%
Tired	0%	91.75%	0%	9.18%

Accuracy: Train: 100%, Test: 81.63%

Kappa: -0.0144

3. Random Forest - 2 Classes

Prediction	Truth	
	hungry	other
hungry -	83	12
other -	2	1

Sensitivity	Specificity	Precision	Misclassification
87.37%	33.33%	97.65%	14.29%

Accuracy: Train: 100%, Test: 85.71%
Kappa: 0.0792

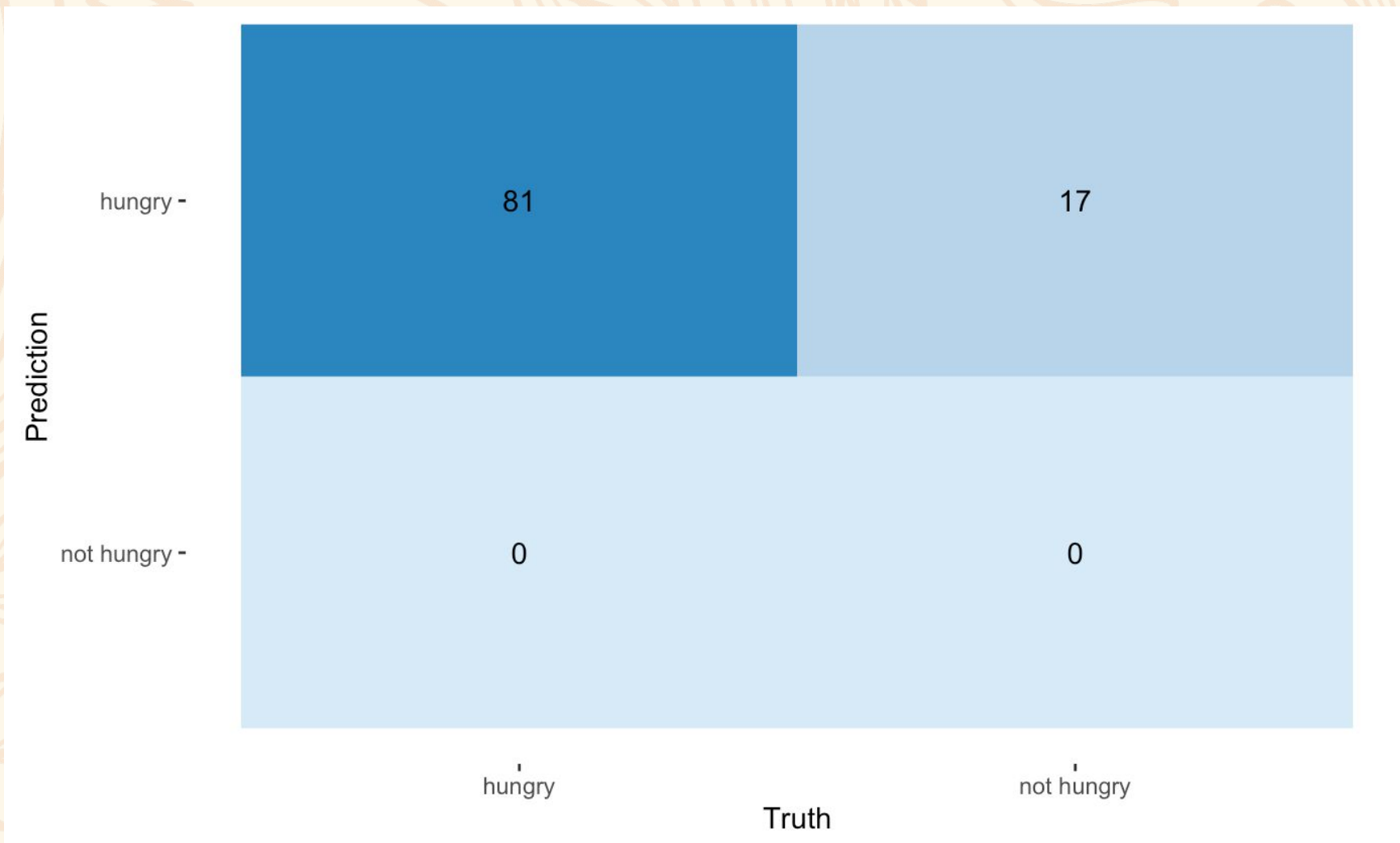
4. Support Vector Machines (SVM) - All Classes

Prediction						
	bellypain -	0	0	0	0	0
	burping -	0	0	0	0	0
	discomfort -	0	0	0	0	0
	hungry -	6	1	8	73	10
	tired -	0	0	0	0	0
		bellypain	burping	discomfort	hungry	tired
		Truth				

	Sensitivity	Specificity	Precision	Misclassification
Belly	0%	100%	NaN	93.88%
Burping	0%	100%	NaN	98.98%
Discomfort	0%	100%	NaN	91.84%
Hungry	100%	0%	79.49%	NaN
Tired	0%	100%	NaN	89.80%

Testing Accuracy: 74.49%
Kappa: 0

4. Support Vector Machines (SVM) - 2 Classes



Sensitivity	Specificity	Precision	Misclassification
100%	0%	82.65%	NaN

Testing Accuracy: 82.65%
Kappa: 0

Summary

<u>Model</u>	<u>Testing Accuracy - All Classes</u>	<u>Testing Accuracy - 2 Classes</u>
Logistic	31.58%	73.47%
KNN	80.95%	84.69%
Random Forest	81.63%	85.71%
SVM	74.49%	82.65%



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Bad Statistics

Including junk cries in models

<u>Model</u>	<u>Testing Accuracy</u>
Logistic Regression	83.82%
KNN	84.78%
Random Forest	81.02%
SVM	39.86%

Modelling parameters on entire data (including test set)

<u>Model</u>	<u>Testing Accuracy</u>
Logistic Regression	85.26%
KNN	84.25%
Random Forest	100%
SVM	89.34%

Feature Selection Cherry Picking

<u>Model</u>	<u>Testing Accuracy</u>
Logistic Regression	83.16%
KNN	83.61%
Random Forest	74.45%
SVM	86.23%



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Conclusions

Conclusions

- It is important to acknowledge that the dataset is skewed heavily
- 'hungry' cries are in a dominate place over all other types of cries which led to bias and hard for the models to accurately classify the data
- KNN generally works better in less noisy data and a smaller dataset. Maybe downsampling the data would improve the accuracy of these models, and also including more data of different types of cries.
- Random Forest seems to be the best model from the good statisticians' perspective as it archives the accuracy of 85.71%.
- Overall, it is incredibly hard to tell whether the models we ran were accurate or not due to the skewness of the data. This meant that even reducing the number of variables or number of classes to be predicted did not improve testing accuracy for some of the models.

Recommendations

- Include more observations to make a larger dataset and improve balance of data to properly train models into predicting classes other than hungry cries
- Make sure to reduce dimensionality of the data since PCA is important in choosing the factors that account for most of the variance in the data, especially in this kind of data where the number of predictor variables are massive and makes the data incredibly noisy. It is good to understand which variables are useless to reduce the runtime for models
- Deeper understanding of the meaning of all various variables of acoustics features in the dataset and try to make more interactions in between.
- Though using the unfiltered data as we did from the bad statistician's' perspective, we achieved the highest accuracy ever (100%), we still have to be honest with clients and our own research goal by cleaning the dataset thoroughly.